



Creating Panoramic Images Using ORB Feature Detection and RANSAC-based Image Alignment

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Abstract

This paper details the development and execution of a project to combine a series of horizontally overlapping photographs into a single panoramic image. The process involves the use of the Oriented FAST and Rotated BRIEF (ORB) feature detector and descriptor, and the Random Sample Consensus (RANSAC) algorithm for automatic image alignment. The final panoramic image is created by blending the images together seamlessly. Two methods for stitching the panorama are discussed: using translations with pre-spherically-warped input images, and using homographies to align the input images directly. Aiming at the problems of large scale rotation registration error, low registration rate and strong randomness and instability of random sampling consistency algorithm in the Rotational invariance binary description algorithm, a fast feature matching algorithm combining ORB and RANSAC is proposed. First, the feature point extraction method is optimized to eliminate the influence of feature edges. Then a simplified pyramid scale space model is constructed to improve the scale space structure of layered images, Reduce the number and layers of generated images; Then the gradient direction is used to improve the main direction extraction mode in the traditional ORB algorithm to improve the accuracy of the main direction of feature corners. Finally, the RANSAC algorithm is improved by constructing a block random sampling detection method to improve the stability of the RANSAC algorithm and the accuracy of Image registration.

Keywords

Panoramic Images, ORB Feature Detection, RANSAC, Image Alignment

1. Introduction

The allure of panoramic imagery lies in its ability to capture wide-angle views that are difficult to obtain with standard photographic equipment. However, creating panoramic images from a series of overlapping photographs is a complex task that requires a robust solution to several interdependent problems. These include the detection and matching of features across images, accurately aligning overlapping areas, and finally blending the images in a seamless manner.

In this project, we employ the ORB feature detector and descriptor to identify robust, discriminative features in the images that facilitate the matching process. Once these features are detected and matched, we use the RANSAC algorithm to automatically align the photographs by determining their overlap and relative positions. Lastly, the images are blended to create a seamless panoramic image.

Our proposed solution forms a pipeline that accomplishes these tasks in a stepwise manner. The steps, visualized using Autostitch UI, operate on the images or intermediate results, eventually producing the final panorama.

Image registration technology is an image processing technology that has developed rapidly in recent years. Image

registration refers to the process of splicing and fusing two or more images acquired by different sensors under different external conditions (different shooting angles, different shooting times, different distortion levels, and different transformation conditions, etc.).

Image stitching technology has a long history of development in fields such as navigation positioning, computer target recognition, remote sensing technology, etc. It has been widely applied in satellite remote sensing technology, disaster monitoring, land and resource investigation, and other fields. Image registration, target change detection and recognition, image mosaic and mosaic are all essential in remote sensing Image registration; In terms of natural disaster monitoring, image registration technology is used to generate panoramic monitoring images to provide accurate data support for task development and project implementation, and provide security for post disaster rescue and reconstruction; In terms of land and resources survey, the use of Image registration technology to generate panoramic remote sensing images has greatly saved the manpower and material expenditure of field [1].

In the research field of Image registration, there are mainly two mainstream Image registration methods: image region based registration method and image feature based registration method. The registration method based on image region directly uses all gray information of the image for registration, which is computationally heavy and has poor effect, and cannot meet the real-time requirements of Image registration. With the rise of image feature processing algorithms in the field of machine vision, image feature based registration methods have attracted widespread attention from experts and scholars due to their low computational complexity, high efficiency, and high accuracy. A large number of excellent registration algorithms have emerged. For example, SIFT algorithm, SURF algorithm, PCA-SIFT algorithm, etc. The literature uses the improved Harris algorithm to detect the feature points with Affine transformation, and uses the local extremum of the normalized derivative (Laplace operator) to represent the feature scale of the local structure, which optimizes the performance of the detector and improves the registration accuracy. Reference improved the registration strategy of the SIFT algorithm in the presence of certain changes in viewpoint, lighting, rotation, scaling, and contrast in the image, and effectively solved the problem of selecting image matching thresholds through RANSAC (Random Sample Consensus) coarse registration. Although the speckle detection method has high accuracy, it requires the establishment of a complex image scale space and the generation of high-dimensional feature descriptors. The computer memory usage is high, the running time is long, and it cannot meet the requirements of real-time performance [2].

This paper outlines the steps and methodologies used to achieve this panoramic stitching, diving into the nuances of feature detection and matching, image alignment, blending, and finally, cropping for a perfect panoramic output.

2. Comparison of Image Alignment Methods

Image alignment, also known as image registration, is a critical step in the process of creating a panoramic image. The objective of image alignment is to determine the geometric transformation that best aligns the corresponding points in the overlapping areas of two or more images. There are several methods available for image alignment, each with its own advantages and disadvantages. Here, we'll discuss several of these methods including Direct Linear Transform, Affine Transform, Projective Transform, and the method we used in this project, Homography and RANSAC, to highlight the variety of techniques available and to compare their relative strengths and weaknesses.

2.1 Direct Linear Transform (DLT)

DLT is a common technique used in computer vision and photo grammetry for solving sets of equations to find unknowns, such as the coefficients of a transformation matrix. This technique requires at least 6 corresponding points in two images to calculate the homography matrix.

DLT provides a simple linear solution and works well in many scenarios. However, it's susceptible to noise and outliers in the corresponding point sets, which can lead to significant errors in the estimated transformation.

2.2 Affine Transform

An affine transform preserves lines and parallelism in geometry. It models transformations including scale, rotation, shear, and translation. Affine transform requires at least 3 corresponding points in two images.

This method is computationally less expensive than the projective transform and the homography methods, and can be a good approximation when the scene is nearly planar or the images are taken from a small field of view. However, the affine transform cannot model perspective distortion which is common in wide field-of-view images,

hence limiting its usage in panoramic stitching [3].

2.3 Projective Transform (also known as Homography)

A projective transform, or homography, is the most flexible transformation and can model scale, rotation, shear, translation, and perspective distortion. This method requires at least 4 corresponding points in two images.

Projective transform is useful when dealing with images taken from different viewpoints or with different camera orientations. It works well even in the presence of perspective distortion, which makes it particularly useful for panoramic image stitching. However, it's more computationally expensive compared to affine and DLT due to the complexity of the transformation.

2.4 Homography and RANSAC

In our project, we used homography in conjunction with RANSAC (RANdom SAmple Consensus) for image alignment. RANSAC is an iterative method used for robust fitting of a model in the presence of many data outliers. In the context of our project, it's used to find the best-fit homography matrix that maps the feature points from one image to the corresponding feature points in another image.

This combination is effective because homography can model the complex transformations between images, and RANSAC can deal robustly with outliers in feature matches. Despite being computationally expensive, this method offers excellent performance in terms of handling different viewpoints, camera orientations, and outlier robustness, making it well-suited for creating panoramic images.

In conclusion, the choice of the image alignment method depends largely on the specific requirements of the project. DLT offers simplicity, Affine is computationally efficient for smaller fields of view, Projective Transform handles perspective distortions, and the combination of Homography and RANSAC provides robustness against outliers. It's important to consider the nature of the images, computational constraints, and the presence of outliers when choosing the most suitable image alignment method [4].

3. Methodology

The methodology for creating a panoramic image can be divided into several distinct steps.

Step 1: Capture Images

The initial step in the process of creating a panoramic image is obtaining a set of images that can be seamlessly stitched together. This requires capturing a series of photographs where each image has substantial overlap with its adjacent images. Overlap ensures that there are sufficient common features that can be detected and matched across images.

In this project, images were captured both with the aid of a tripod and handheld. Utilizing a tripod guarantees stability and consistency across the series of photographs, which is beneficial when capturing images for a panorama. However, handheld images are also viable and increase the flexibility of capturing panoramas in a wider range of environments.

Step 2: Spherical Warping

For certain methods of stitching panoramas, the input images need to be preprocessed into a different coordinate system. Specifically, when using translations to stitch the panorama, the images need to be pre-spherically warped.

Spherical warping involves transforming the image such that it appears as though it has been projected onto a sphere. This can help mitigate distortions caused by wide-angle lenses or perspective differences between images.

The compute Spherical Warp Mappings routine was employed for this purpose. The routine converts the given spherical image coordinate into the corresponding planar image coordinate, following the coordinate transformation equation from the lecture notes. It further applies radial distortion using the respective equation from the lecture notes.

It is worth noting that spherical warping is an optional step, primarily used when translating the images for panorama creation. When using homographies, the input images can be directly aligned.

Step 3: Feature Extraction

Feature extraction is a critical step in image stitching. Features can be defined as patterns or unique characteristics that occur in an image. Common features can include edges, corners, or blobs of color. These features act as points of correspondence in different images, allowing for their alignment.

For this project, we chose to use the ORB (Oriented FAST and Rotated BRIEF) feature detector. ORB is a fusion of the FAST key point detector and the BRIEF descriptor with several modifications to enhance performance. ORB is invariant to rotations and resistant to noise, making it suitable for our purpose.

The extracted features are stored for each image, providing a basis for the subsequent matching process.

Step 4: Feature Matching

Once features have been detected in each image, the next step is to match these features across the images. This involves comparing the detected features and identifying pairs of similar features.

The feature matching process is essential to establish common points of reference between the images. These matched features provide a means of aligning the images relative to each other.

The ORB descriptor, which provides a binary description of the features, was used for this matching process. This descriptor allows for a fast and efficient matching process, significantly reducing the computational resources required.

Step 5: Image Alignment Using RANSAC

With the matched features established, the next step is aligning the images. Image alignment is the process of determining the transformation that maps points in one image to corresponding points in another image. In the context of panoramic stitching, image alignment involves estimating the translations and homographies that best align neighboring image pairs.

The RANSAC (Random Sample Consensus) algorithm was employed for this purpose. RANSAC is a robust method for fitting a model to data with a high degree of noise or outliers. It works by randomly selecting a subset of the data, fitting a model, and then measuring the quality of the fit with a scoring function. This process is repeated numerous times, and the model with the highest score is selected.

The align Pair routine provided in the alignment.py file implements this functionality. The routine takes two feature sets, a list of feature matches, and a motion model as parameters, and then estimates the inter-image transform matrix. It uses RANSAC to estimate this transform, first selecting a minimal set of feature matches, estimating the corresponding motion (alignment), and then using the get Inliers function to identify the matches that agree with the current motion estimate [5].

Step 6: Drift Correction

Image alignment can introduce an issue known as drift, where the alignment of images gradually shifts as more images are added to the panorama. This is particularly noticeable in panoramas covering a large field of view or a full 360 degrees.

Drift correction is hence an essential step in the stitching pipeline. It involves correcting the alignment of the images so that the first and last image aligns correctly with each other.

This was accomplished using a least squares method. The Least Squares Fit routine takes in the matches estimated as inliers and computes a least squares estimate for the translation or homography. The resulting transform is then applied to correct the drift.

Step 7: Image Blending

Once the images have been accurately aligned, the next step is to blend them together to create the final panoramic image. Image blending involves overlaying the aligned images and combining their pixel values in a manner that creates a smooth transition from one image to the next.

The image blending process was conducted in the blend.py file using routines like image Bounding Box, blend Images, accumulate Blend, and normalize Blend. The image Bounding Box routine determines the box bounding the image after applying the homography. The get Acc Size routine is used to figure out how large the final stitched image will be and their absolute displacements in the panorama. The accumulate Blend routine resamples each image to its final location and blends it with its neighbors.

A simple feathering function was used as the weighting function to create smooth transitions between images. Feathering reduces the weight of an image's contribution as the distance from the image center increases, resulting in a gradual transition that reduces visible seams.

Step 8: Cropping and Finalization

The final step in the stitching process is cropping the result to create the desired panoramic view and bringing the final image into a viewer. It's important to ensure that the resulting panoramic image fits the viewer's aspect ratio and provides the intended wide-angle view.

The compute Spherical Warp Mappings routine also converts the panoramic image to a planar image coordinate

system, providing the inverse map to warp the image. The transformation is also applied to correct any radial distortion.

The result is a panoramic image that seamlessly stitches the input images, presenting a broad, unbroken view of the captured scene.

4. Discussion and Conclusion

This project has provided a deep insight into the complexities of panorama creation. From capturing the images, preprocessing them via spherical warping, feature detection and matching, alignment via RANSAC, to the final blending, each step contributes crucially to the outcome. Despite the challenges involved, successful execution results in an aesthetically pleasing and immersive viewing experience that standard photographs cannot offer.

This approach can further be optimized and enhanced by introducing multi-band blending, exposure compensation, or HDR techniques. These additions will improve the quality of the panoramic images and make them more visually appealing [6].

The project has been a valuable exercise in understanding how feature detection algorithms like ORB, robust fitting algorithms like RANSAC, and blending techniques can be used in conjunction to create panoramic images. The panoramic stitching pipeline created in this project can be used as a base for further research and development in this field.

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